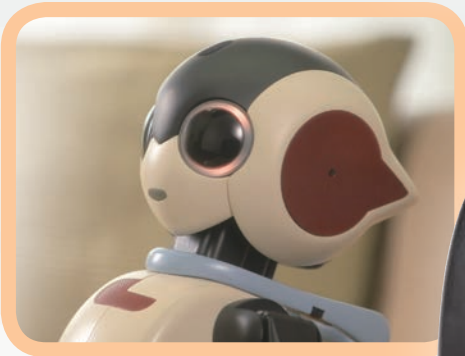


Build your own Robi

Pack **17**



Stages 63-66:
Begin wiring
Robi's head

Build your own **Robi**

CONTENTS

Assembly Guide

p352

Stage 63: Prepare Robi's second eye sensor **p352**

Stage 64: Fit the eye sensor cables **p355**

Stage 65: Mount the LED mouth board **p357**

Stage 66: Rebuild Robi's head and test his movement **p360**

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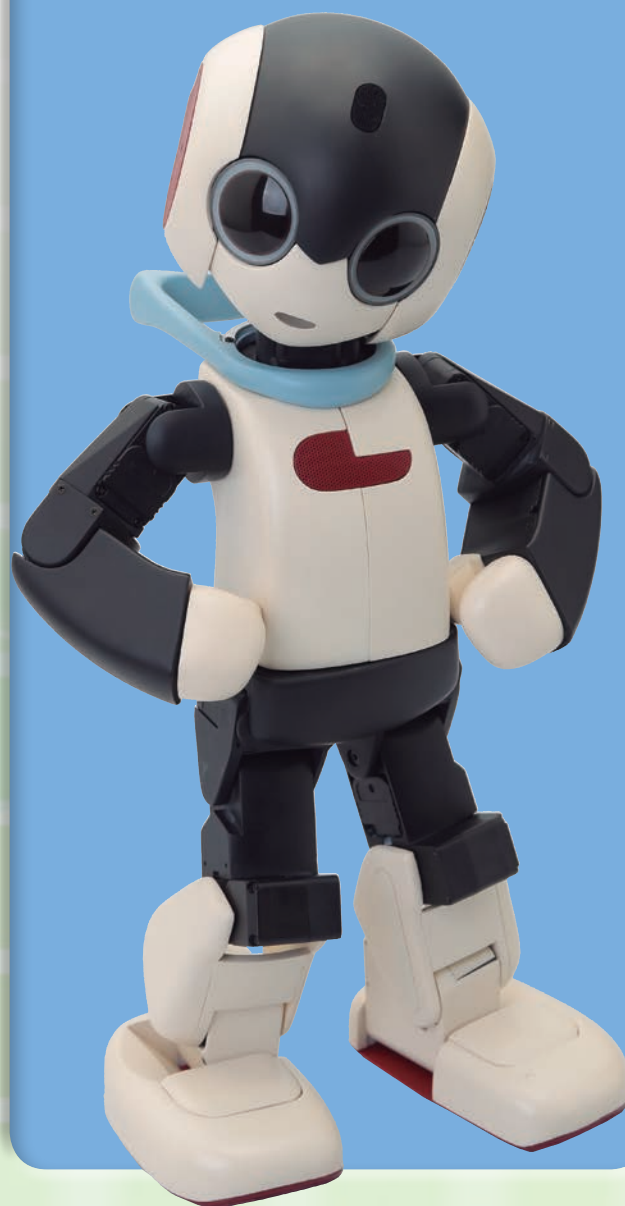
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STAGE 63: PREPARE ROBI'S SECOND EYE SENSOR

Following on from the work in the last pack, in this session you will prepare the second of Robi's infrared eye sensors.

In this stage, you will fit the second of Robi's infrared eye sensors. The work is the same as for when you fitted the first

sensor, but, as ever, always follow the instructions closely, especially when handling exposed electronics parts.

PARTS TO BE ASSEMBLED



1



YOUR PARTS

1 Infrared eye sensor

SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

M2 x 5mm pan-head screws x 3 (Stage 62)



*Silver stickers
(Stage 62)*



*Eye sensor frames
(Stage 62)*



*Eye without sensor
(removed Stage 62)*

FITTING THE SILVER STICKERS



- 1** Take the eye that you did not fit with a sensor in the previous pack, and gently push its black iris out from its socket.



- 2** Look inside the eye socket and locate the two screw holes outlined in red above. The silver stickers supplied in this stage will go over these areas, to ensure that a solid ring of silver lines the eye socket.



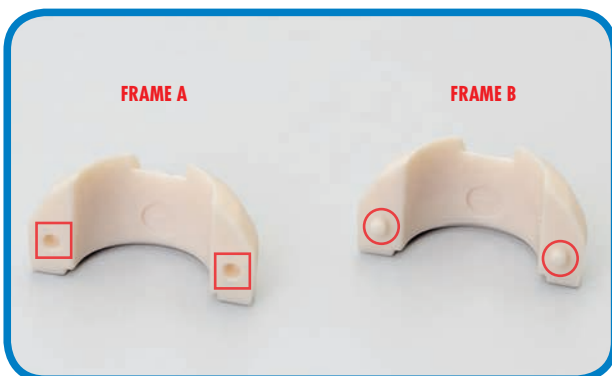
- 3** Fit a silver pad over each of these areas. Using tweezers or a toothpick will help if you find it tricky to do with your fingertips.



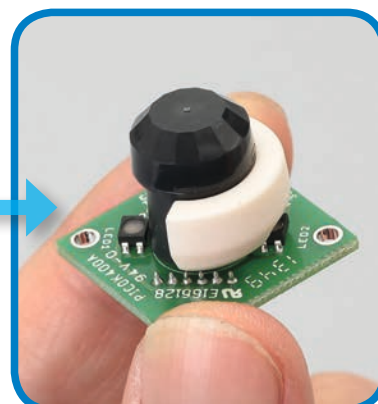
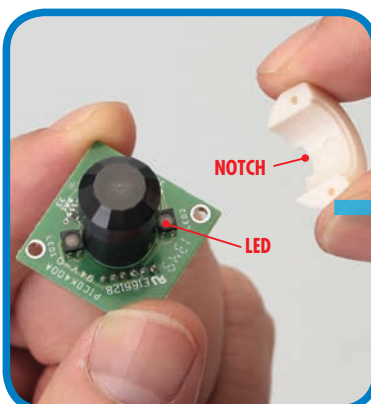
- 4** Your assembly should look like this, with the stickers creating a solid ring of silver that lines the inside of the eye socket.

THE EYE SENSOR FRAMES

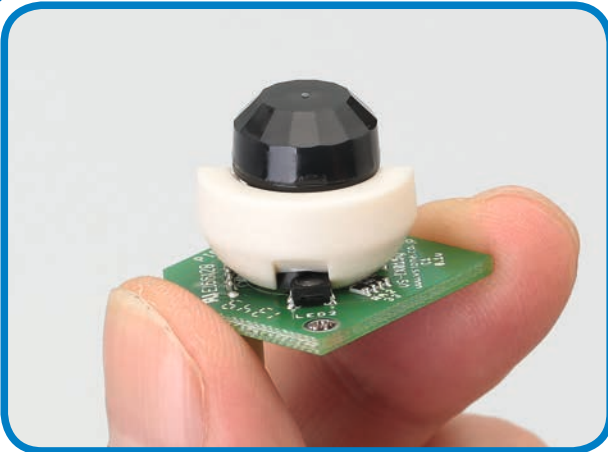
Only hold the exposed circuit board of the eye sensor by its sides!



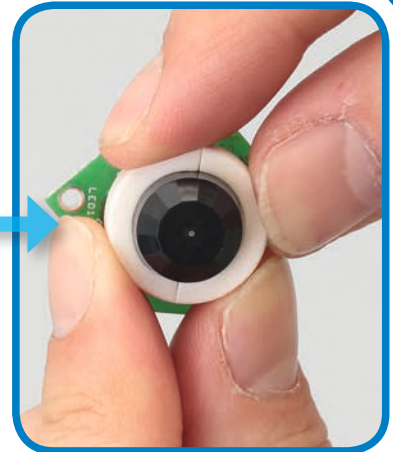
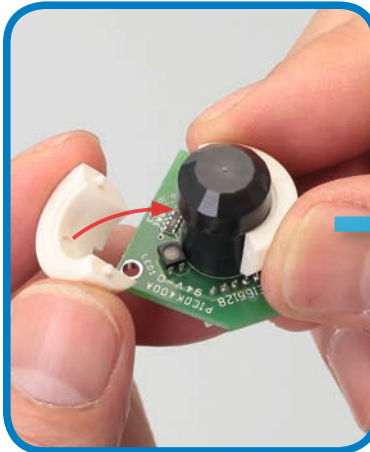
- 5** Familiarise yourself with the eye sensor frames. Each comes in two parts, A and B. Frames A have holes on their ends, which will hold the locating pins of frames B, as seen above.



- 6** Take a frame A and fit it onto the base of the infrared eye sensor supplied in this stage. The frame's notch will accommodate one of the sensor's LEDs, so align the part exactly as shown.



7 Your assembly should now look like this.

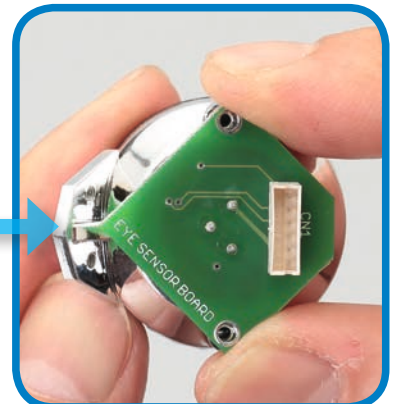
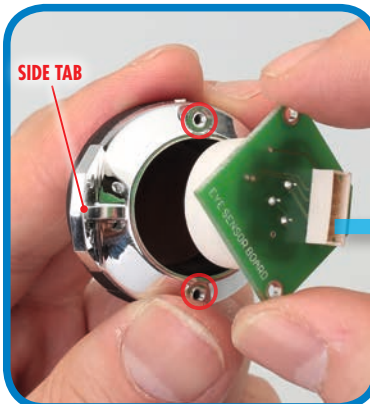


8 Add a frame B to the assembly as shown, with its locating pins fitting into the corresponding holes of frame A. Press the parts firmly together.

FIT THE EYE SENSOR



9 Replace the iris removed from its socket in Step 1.



10 Carefully fit the sensor into the socket. The two screw holes in the sensor's backing should align with those of the socket, with its uncut corner next to the socket's side tab.



11 Use two of the M2 x 5mm screws saved from Stage 62 to secure the sensor to the eye socket.

Assembled eye sensor



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Storing the parts

Store the delicate eye sensor assembly in a safe place where it is protected from knocks, scratches and static electricity.



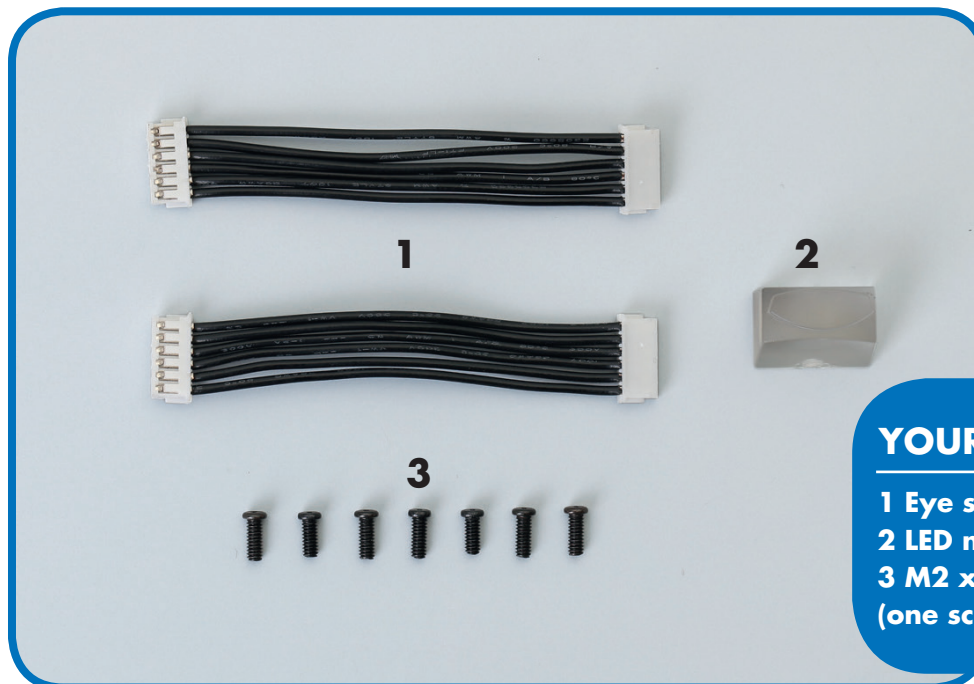
STAGE 64: FIT THE EYE SENSOR CABLES

Now that both eyes are fitted with their infrared movement sensors, you can link up the cables that will connect them to the rest of Robi's electronic systems.

Robi's eye sensors and LEDs are connected up to the rest of his systems – and ultimately the CPU – via two six-strand cables, supplied with this stage. The work in this assembly is very

straightforward, though you must be sure to follow the steps closely to fit the cables' connectors the right way round, or you risk damaging their delicate metal pins.

PARTS TO BE ASSEMBLED



YOUR PARTS

- 1 Eye sensor cables x 2
- 2 LED mouth block
- 3 M2 x 5mm pan-head screws x 7 (one screw is a spare)

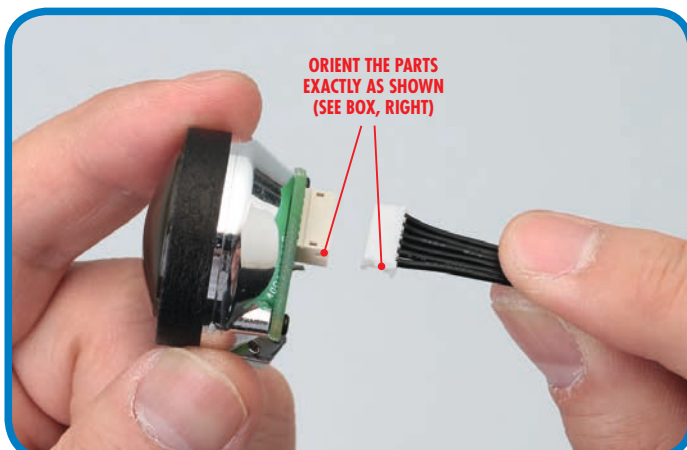
SAVED PARTS

You will be using parts from the previous stages, so be sure to have these to hand before beginning.

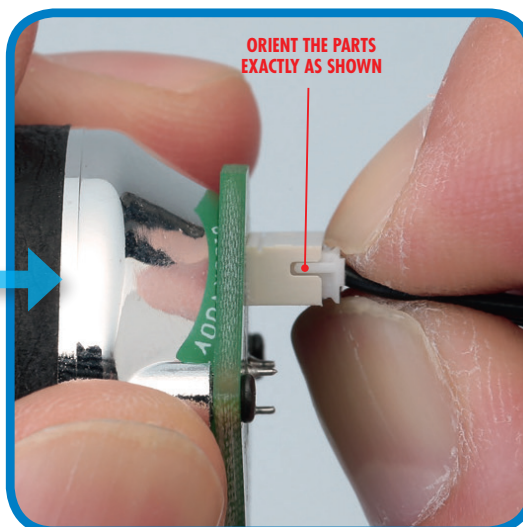
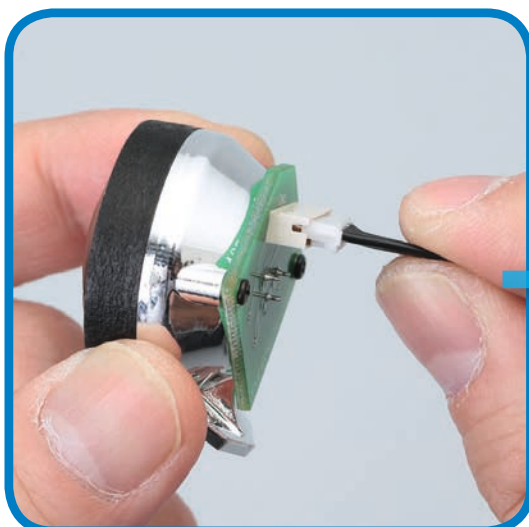


*Eyes with sensors fitted
(Stage 62 and 63)*

FIT THE EYE SENSOR CABLES



- 1 Take the first eye and fit the connector at the end of one of the eye sensor cables into its socket, oriented exactly as shown. There is a small ridge on the side of the connector's housing to help you slot it in correctly.



Use the ridge and groove to help ensure the connector fits into its socket correctly!

- 2 Repeat Step 1 to fit the second eye sensor cable to the other eye, again being sure to fit the parts as shown.



Assembled eyes

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

Connecting the eye sensor cables

Robi's eye sensors are linked up to the CPU via the LED mouth board (fitted in the next stage) using two six-strand cables, each of which has a white plastic connector at its ends. On the inside of these are six metal pins that, when pushed into the eye sensor and mouth board sockets, connect the circuit. As the pins are set only on one side of each connector's housing, it is important you orient these as shown in the step photos.



Storing the parts

Store the delicate eye sensor assembly in a safe place where it is protected from knocks, scratches and static electricity.



STAGE 65: MOUNT THE LED MOUTH BOARD

In this stage, you will connect up the infrared sensor cables, fitted to the eye in the previous stage, to the LED mouth board.

The LED mouth board supplied in this stage has two functions: firstly, it holds the LED that will light up Robi's mouth when he talks; and secondly, it links the infrared eye sensors and, later, voice-recognition board, to the rest of Robi's circuitry. Just as when you fitted the

sensor cables to the eyes themselves, you must take care when fitting the cable connectors. And, as ever, great care must be taken when you are handling exposed electronic components such as those on the LED mouth board!

PARTS TO BE ASSEMBLED



1



YOUR PARTS

1 LED mouth board

SAVED PARTS



Eyes (Stage 64)



M2 x 5mm pan-head screws x 2 (Stage 64)

LED mouth block (Stage 64)



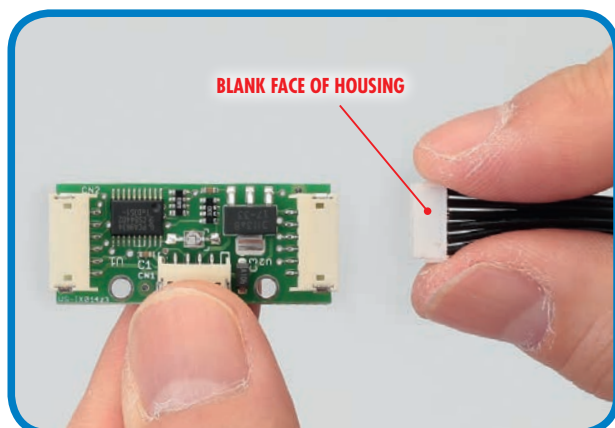
Robi's face (removed in Stage 62)



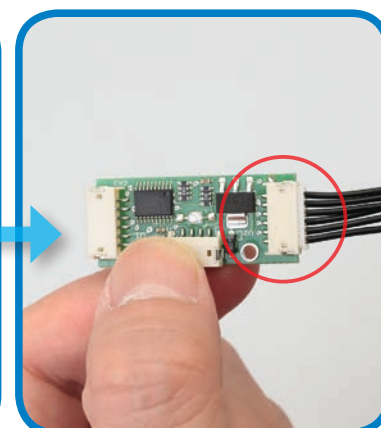
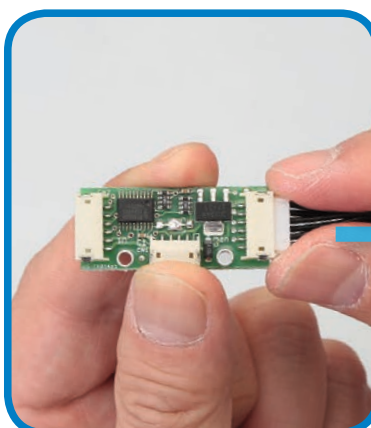
Protective pads (Stage 3)



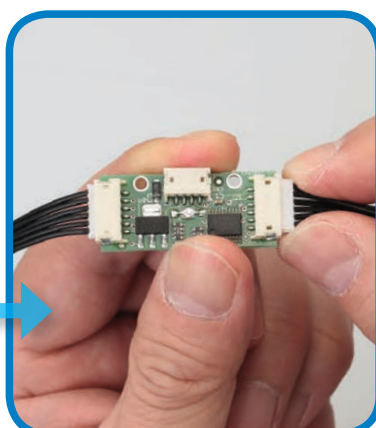
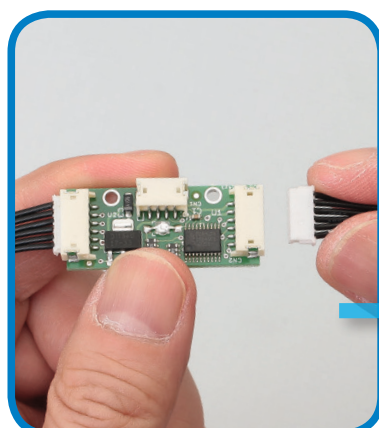
LINK UP THE EYE SENSORS



- 1 Hold the LED mouth board as shown (try not to touch any of the exposed electronics on its face) and line up the free connector leading from one of the eyes prepared in the previous stage. Make sure the blank face of the connector's housing – where the internal metal pins are not visible – is facing you.



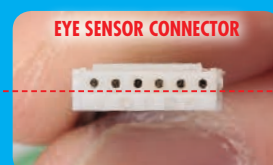
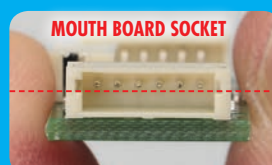
- 2 Slot the connector into the right-hand socket, making sure it is the right way round and is set securely. Again, be careful with the exposed components on the board – if you need better grip, touch only the plastic socket housings, as shown above right.



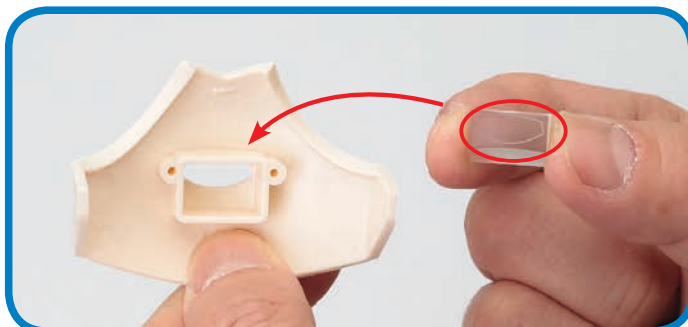
- 3 Repeat Steps 1 and 2 to connect the other eye to the board, again making absolutely sure that you set the connector correctly.

Connecting the eye sensors

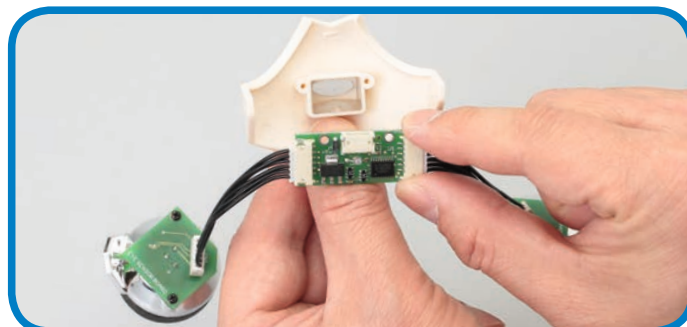
The eye sensors are linked to the CPU via the LED mouth board supplied with this stage. Just as when you connected the eye sensor cables to the eyes themselves, when joining these to the mouth board you must ensure that they are oriented the correct way round. The photos below illustrate how the parts fit.



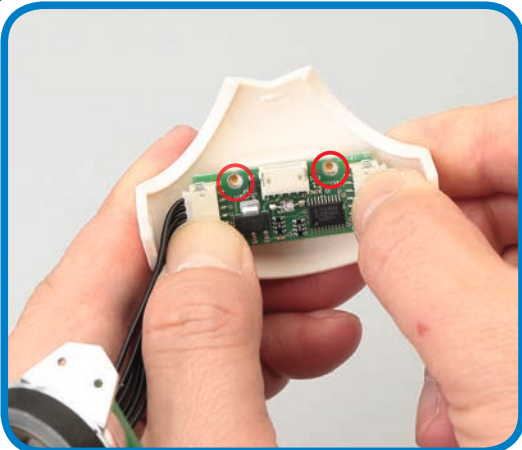
FIT THE LED MOUTH BLOCK AND BOARD



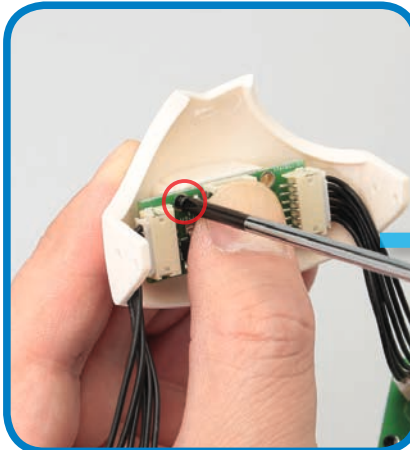
- 4 Line up the LED mouth block to the inside of the face panel, removed in Stage 62. The raised, shaped section will sit in the correspondingly shaped hole.



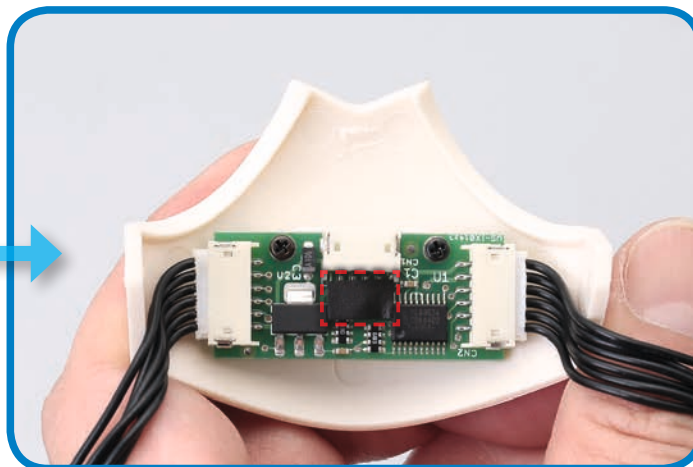
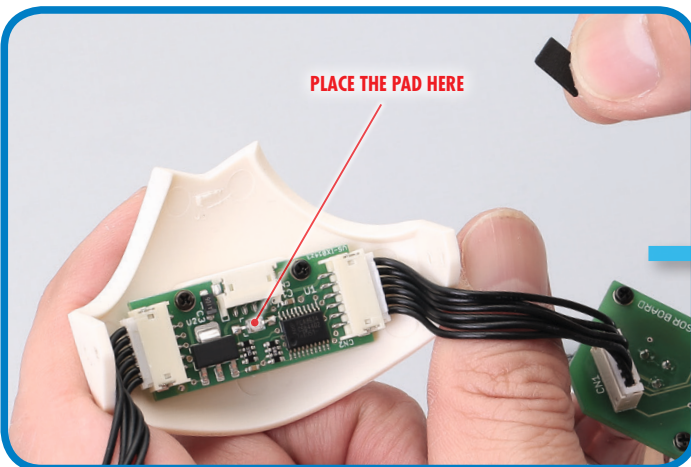
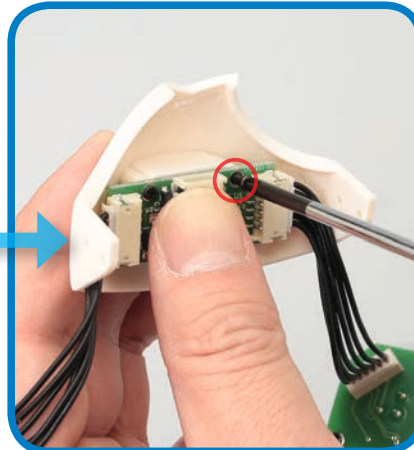
- 5 Now line up the LED mouth board to the block, as shown. Make sure the eyes remain in place.



- 6** Position the mouth board so that the two circled holes line up with those on the inside of the face panel.



- 7** Use the two M2 x 5mm pan-head screws saved from Stage 64 to secure the board.



- 8** Take a protective pad and fit this over the rear of the LED, in the centre of the board. This will prevent light filtering into the inside of Robi's head assembly.



Assembled eyes and face

Storing the parts

As with any of Robi's parts that include electronic components, store this stage's assembly in a cool, dry place – preferably in a sealed or padded bag where it will be protected from scrapes, knocks and static.

Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

STAGE 66: REBUILD ROBI'S HEAD AND TEST HIS MOVEMENT

In this stage, you will set Robi's eyes (now fitted with their sensors) back into his head, before performing another movement test.

After dismantling Robi's head in order to fit his eyes with their infrared movement sensors, you are now ready to reassemble the head and proceed with the build. Before this, though, you will be running another movement test

using the battery and CPU, as you have done before. This will be the final movement test to check just one section of Robi's body – the next one will be the final, full-body test before completion in the next pack!

PARTS TO BE ASSEMBLED



1

2



YOUR PARTS

- 1 LED mouth board cable 11cm
- 2 VR board cable 6cm

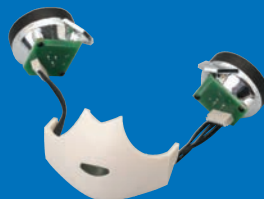
SAVED PARTS



Battery (Stage 38)



Power distribution board
(Stage 40)



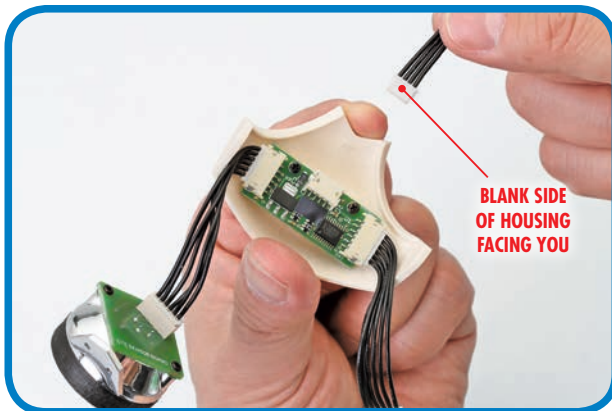
Robi's face and eyes
(Stage 65)

M2 x 4.5mm
countersunk screws x 8
(removed Stage 62)

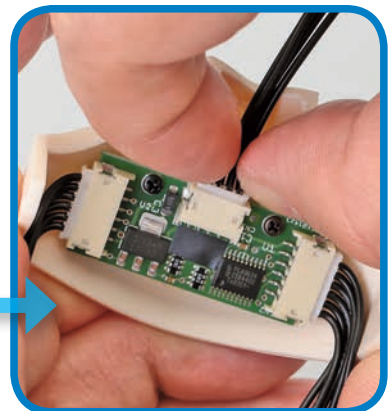
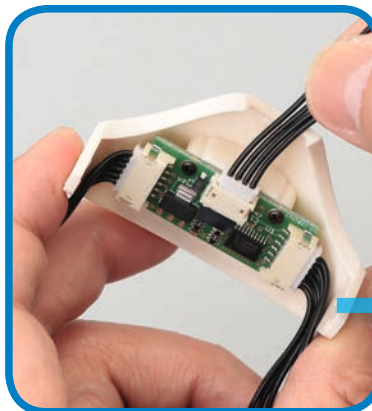


Body and head assembly
(Stage 62)

ADD THE SENSOR/LED CABLE

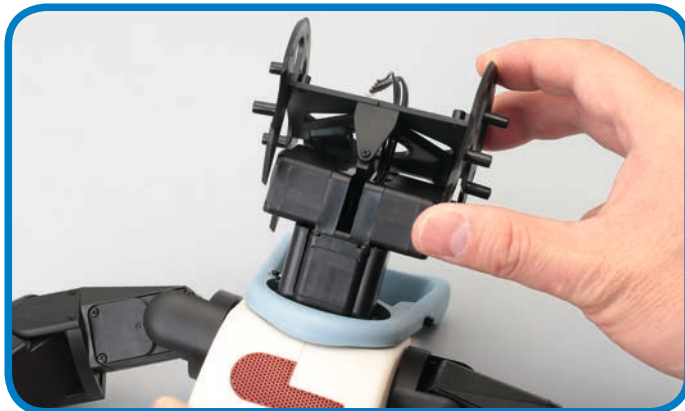


- 1** Line up the 11cm LED mouth board cable to the lower section of the face assembly. The blank rear face of the connector's housing (where you cannot see the metal pins) should face you.

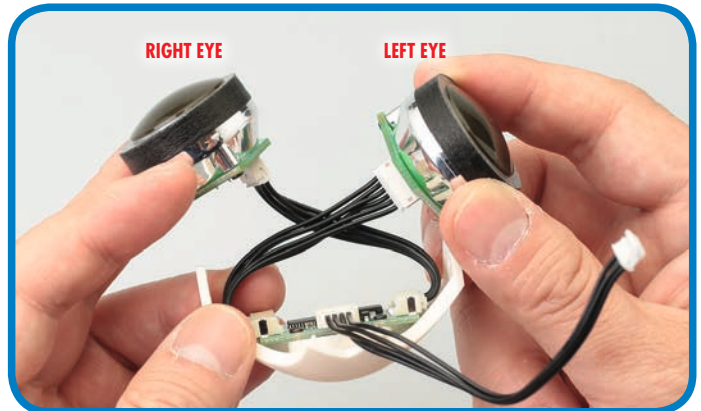


- 2** Slot this securely into the free socket at the bottom of the LED mouth board.

FIT THE EYES



- 3** Take the body and head assembly and tilt the head frames upward a little by hand.



- 4** Cross the two eyes over one another. This is the final position they will be set to.

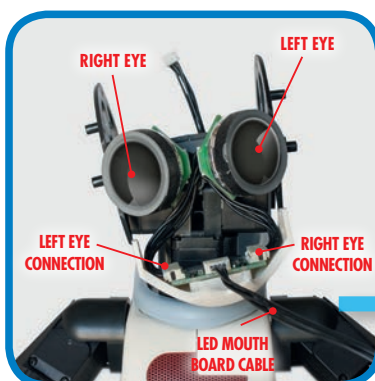


- 5** Line up the left eye to the left head frame. Match the arrowed and circled holes on the eye socket and frame, then use two M2 x 4.5mm countersunk screws, saved from Stage 62, to secure the parts.



- 6** Repeat for the right eye.

ATTACH THE FACE



7 Confirm that the connections of the infrared sensors and the eyes that house them are on opposite sides, as shown. Then pull the LED mouth board cable upwards between the eyes and place the face in position beneath them.

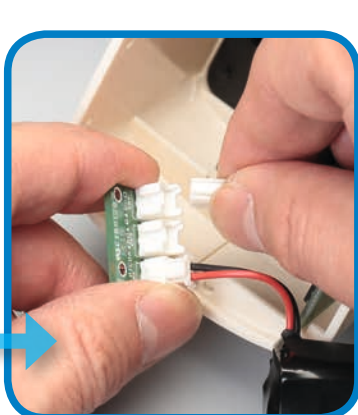
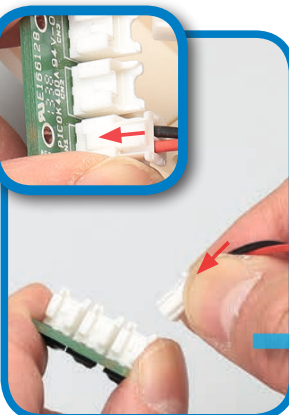
8 Line up the face to the left side of the head frame, so as to align the circled holes on both parts.



9 Insert and tighten two M2 x 4.5mm countersunk screws, saved from Stage 62, into these holes.

10 Repeat for the corresponding two holes on the right side of the face and the head frame.

TEST THE BODY AND HEAD



11 Fit the (charged) battery's cable into socket CN1 on the power distribution board. Then, with the scarf switch in the OFF position, fit Robi's power/switch cable into CN3.

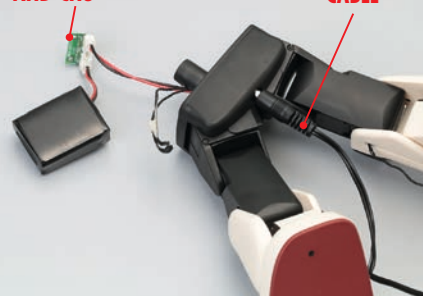
RECHARGING THE BATTERY

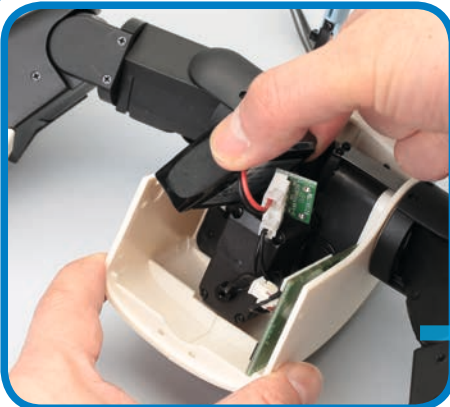


To recharge Robi's battery, connect it and the power cord to the power distribution board, in sockets CN1 and CN3 respectively, then plug the mains charger into the back of the rear hip panel.

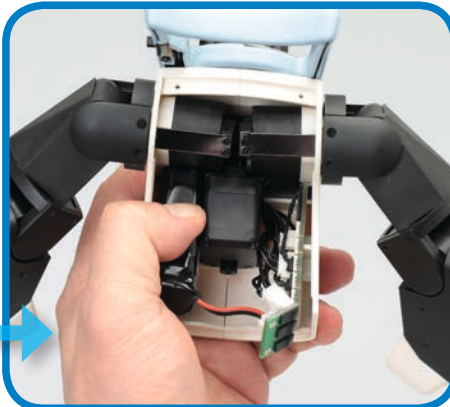
POWER DISTRIBUTION BOARD SOCKETS CN1 AND CN3

CHARGER CABLE





12 With the cables still connected, slot the battery into the space next to the back servo on the left inside of Robi's body. Then pick Robi up, holding the battery in place.



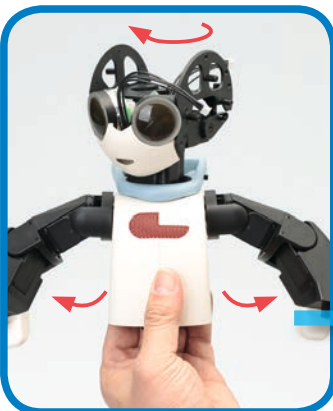
13 Turn the power ON.



14 As soon as the power is switched ON, Robi's servos will return to their starting/neutral positions, which are shown above from the front and side. NOTE: If your Robi does not look like the one above when in its neutral position, see the troubleshooting guide later in this stage.



15 The first phase of the test will see Robi's forearms extend and his head tilt upwards, before both return to their starting positions, shown in the previous step.



16 Next, Robi will turn his head to the right and lift his upper arms outwards, before returning them to their starting positions.



17 Robi's shoulder servos will then lift his arms forward and ahead of him, and, as this happens, the shaft of the back servo will rotate (you will need to view this from underneath to be able to see it clearly).



18 After the movement test is complete, Robi's servos will return to their neutral positions and his speaker will say 'OK!'

19 Once you are happy that everything is in working order, turn Robi OFF and disconnect the power/switch cable from the power distribution board.

CHECK THE NEUTRAL POSITIONS!

Before the tests start, Robi's servos should revert to their neutral positions, so you can check that they are moving as they should be. If, when the neutral positions are reached, you find that your Robi does not resemble the photos in the assembly guide, you will need to address the problem before continuing with the build.

See the troubleshooting guide of this stage to help you re-set Robi's servos.

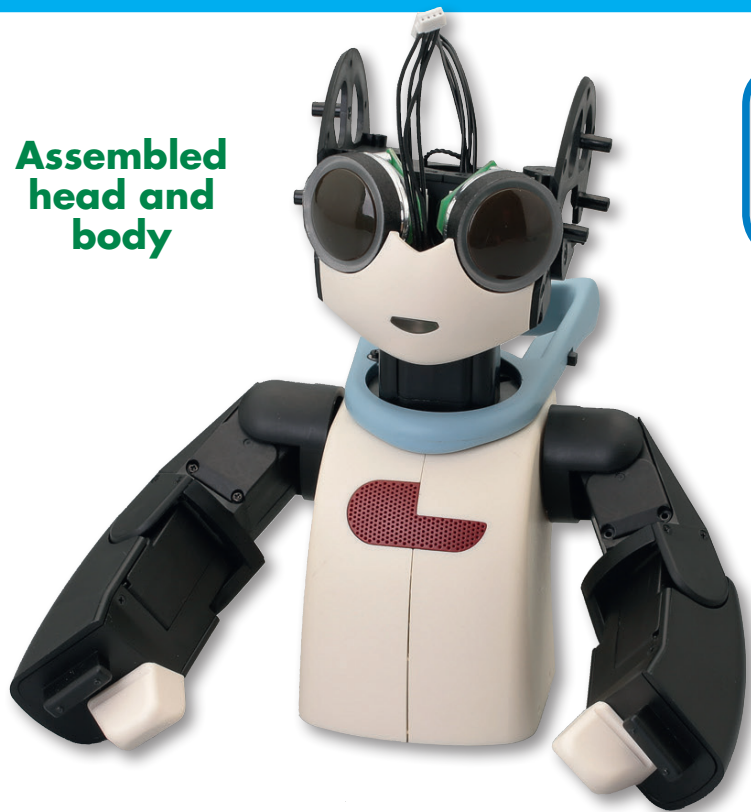


ARMS NOT SET CORRECTLY



HEAD NOT SET CORRECTLY

Assembled head and body



Storing the parts

Keep the shorter, 6cm VR board cable safely until it is needed.



Remember, to see all of Robi's assembly stages, including this one, visit the ModelSpace web page at:

www.model-space.com

TROUBLESHOOTING

In this section, we look again at what to do if the movement test carried out in this stage does not go as it should. The pages that follow focus specifically on what to do if the head or arms are not setting to their starting or neutral positions as they should, in which case it is probable that the D-cut parts joining the components need attention.

ATTENTION!

The photos in the following steps show the left arm, but the instructions, suitably adapted, also apply to the right arm. So if it is the right arm that is faulty, just follow the same basic procedure.

Incorrectly set neutral positions

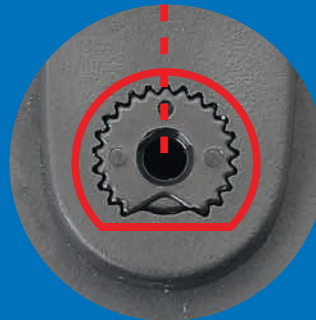
When the test begins, the arms and head should reset to their neutral positions, which are marked in the step photos for each test section. Observe these neutral positions closely from the front and side to assess if they are as they should be. Disconnecting the battery as soon as the parts move into the neutral position will freeze the test, allowing you to inspect them more easily (see below).

If any of the joints does not resemble those shown in the photos, it is probable that it was not set correctly when fitted. To remedy this, you need to check/reset the D-cut joint in question (see box, right). Be careful not to repeat the mistake when you reset the joints.

THE D-CUT



The D-cut components are the key to holding Robi's joints in the correct neutral positions, so they must be set correctly from the start for Robi to function properly! If, when you perform the tests, one or other joint is askew, you will need to dismantle it and reset its joint to the correct position.



The D-cut hole holds the servo shaft straight.



If not set correctly, the joint will be off-centre, and the plastic can be damaged.

IF NONE OF THE SERVOS MOVE

If none of the servos move when the test is started, begin by checking the following points:

1. The battery is fully charged
2. The micro SD card is properly inserted
3. The scarf's switch is ON

If all of the above are OK but the problem remains, disconnect all the servos from the CPU and then test each one individually. Doing this will help you establish which servo group (left arm, right arm, neck/head) is faulty and needs attention.

IF THE SPEAKER DOES NOT SAY 'OK!'

If Robi's speaker does not say 'OK!' at the end of the movement test, check the connection made in Step 15 of Stage 57.

IF ONLY SOME SERVOS MOVE

See the following pages for directions on how to dismantle and reset the problem servos/joints.

IF THE FOREARM IS ASKEW

If you compare your Robi to the photos of the body, head and arms in their neutral positions, and find either of the forearms is set incorrectly (for example touching his body or sticking out too far away from it), you will need to reset the elbow joint.



1 Remove the two circled screws on the problem elbow. **Keep the screws.**

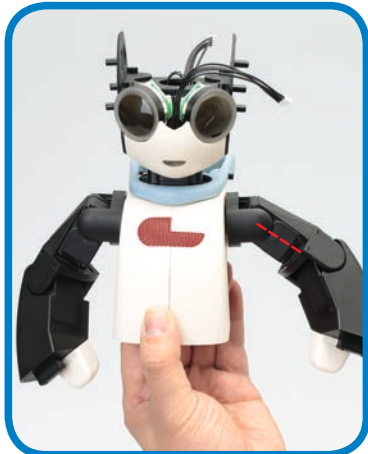


2 Remove the elbow panel, then carefully reset its D-cut socket onto the shaft of the forearm servo. Use the same screws to re-secure the parts.

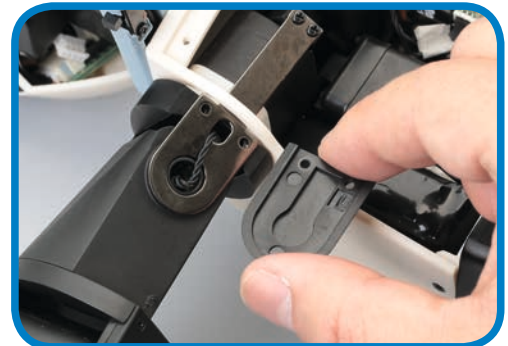
For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's forearms and their servos, see Stages 8 and 9 (right) or Stages 16 and 17 (left).

IF THE UPPER ARM IS ASKEW

If it is one of Robi's upper arms that is set incorrectly, you need to reset the shoulder joint.



3 Remove the two screws holding the shoulder panel in place. **Keep the screws.**



4 Loosen the cable slightly.



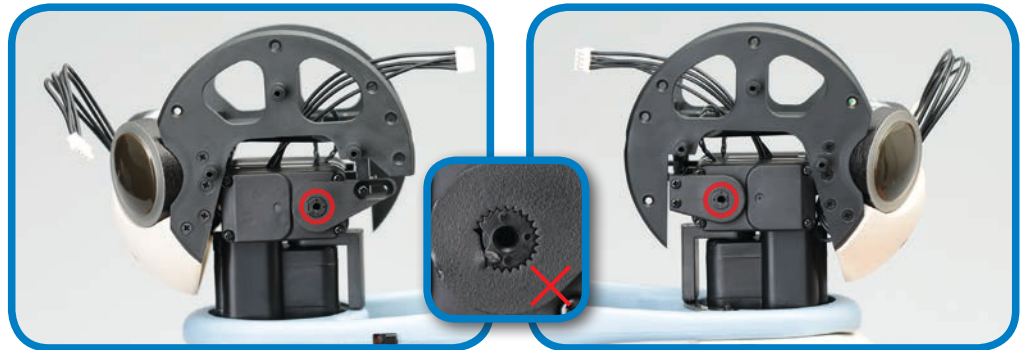
5 Gently remove the upper arm servo, in its frame, from the shoulder, and reset its shaft onto the D-cut in the shoulder frame. Reassemble the parts.



For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's upper arms and their servos, see Stages 11, 12 and 14 (right) or Stages 19, 20 and 22 (left).

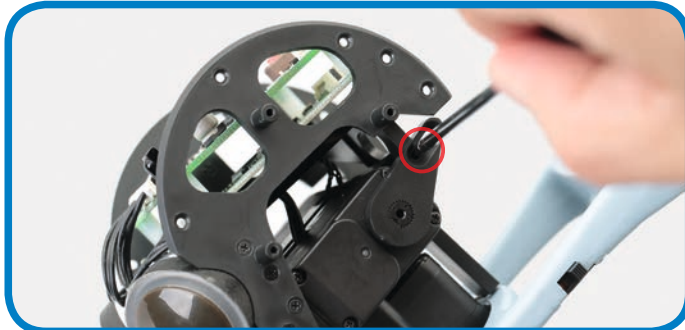
IF THE HEAD IS ASKEW (TILTING TOO MUCH)

If you compare your Robi's head to the photos of the head in its neutral position and find it is tilting either too far forward, back or to either side, you will need to reset the D-cut joints between the head servos and frames. NOTE: if Robi's head is *facing* too far to the left or right, it is the joint of the neck servo that needs attention – see the bottom of this page.



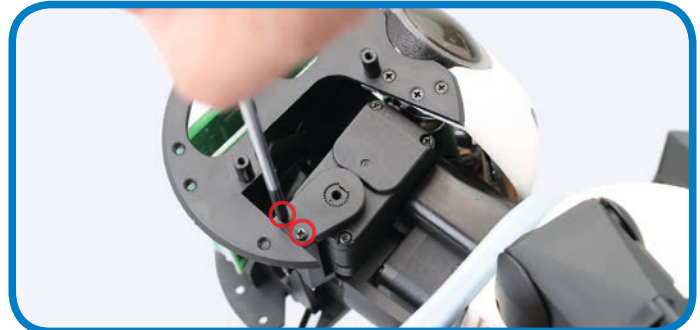
6

The D-cut joint of servo ID=14, on the left side of Robi's head, controls the side-to-side tilt, so reset this to correct any misalignment of the neutral position along this axis. The forward-and-back inclination is set by the joint of servo ID=15, on the right side of Robi's head.



7

To free servo ID=14, in order to remove and reset its servo horn, take out the circled screw. **Keep the screw.**



8

Take out the two circled screws on the right-hand head servo ID=15, to remove and reset its horn. **Keep the screws.**

For further instructions on the process of reassembling and/or resetting the ID numbers of Robi's head and head servos, see Stages 59-61.

IF THE HEAD IS ASKEW (FACING THE WRONG WAY)

If Robi's head is facing too far to the left or right, it is the joint of the neck servo that needs attention.



9

Remove the screws as directed in Steps 7 and 8, then remove the screw circled above. **Keep all the screws, making a note of which ones go where.**



10

Remove the eight screws holding Robi's eyes to their sockets (the four on the left side are circled here).



11 Now remove the circled screw at the front of the head frame assembly to free it.



12 Remove the four screws holding servo ID=14 in its frame, and gently lift it away.



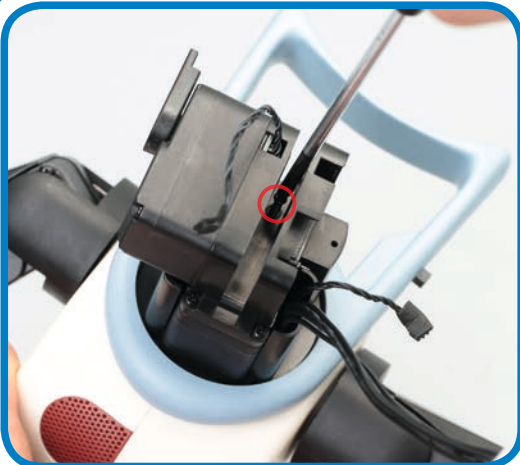
13 Very carefully disconnect both servo cables from their sockets.



14 Now you can unscrew and remove the head lever.



15 Unscrew the neck plate.



16 Remove the screw from the central gap in the head servo frame.



17 Now remove the two circled screws holding the head servo frame to the neck base, and lift it away carefully.



18 You can now access the neck servo and horn, which you should remove and reset as necessary.



For further instructions on the process of reassembling and/or resetting the ID number of Robi's neck and servo, see Stages 54, 57, 59, 60, 61 and 66.

IF THE ARMS ARE ASKEW FROM THE SHOULDER

If either of Robi's arms is not set to its neutral position correctly, you will need to reset the shoulder servo(s).



19 Follow Steps 7-18, then proceed to remove the two screws circled above if you are removing the left arm – the corresponding screws on the right side should be removed if it is the right arm you wish to remove. **Keep the screws, making a note of which ones go where.**





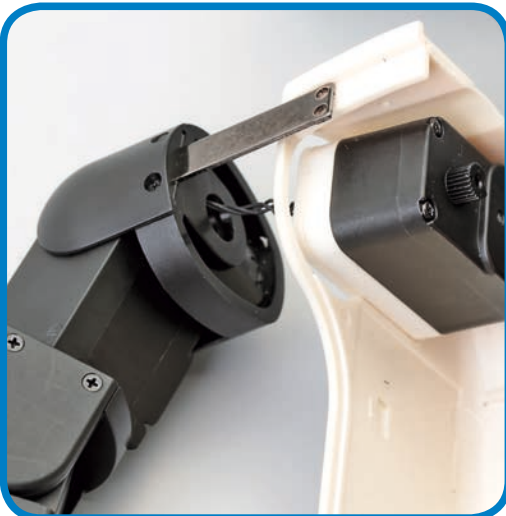
- 20** Very carefully separate the side of the body that houses the problem arm. Do this slowly to avoid any of the cables becoming dislodged or stretched.



- 21** Check the D-cut joint of the shoulder servo, circled above. Be careful not to disturb any of the cables as you do this.



- 22** Remove the two screws holding the metal shoulder joint to the servo horn, then remove the horn.



- 23** You are now able to remove and reset the parts, making absolutely sure that the D-cut joint is set correctly!



At the end of this stage, check again that Robi's arms, neck and head are set to their neutral positions, as shown above.

For further instructions on reassembling and/or resetting the ID numbers of Robi's shoulders, see Stages 14, 55, 56, 57, 59, 60, 61 and 66 (right shoulder) and/or Stages 22, 55, 56, 57, 59, 60, 61 and 66 (left shoulder)

COMING IN PACK 18

Your final **FOUR** complete Stages, as you watch Robi comes to life!

Plus your *Living with Robi* guide

Everything you need to know to get the most out of your new robot companion!



Stage 67

Mount the VR board

PARTS PROVIDED

Robi's voice-recognition board and mounting posts.

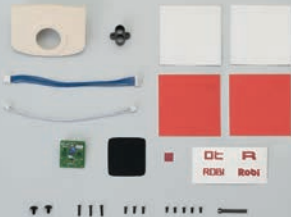


Stage 68

Mount the TV/RC board

PARTS PROVIDED

Robi's TV remote control board.



Stage 69

Set up your Robi's TV remote control capabilities!

PARTS PROVIDED

The infrared LED, cables and more.



Stage 70

Complete your Robi!

PARTS PROVIDED

The final micro SD card and last body coverings.